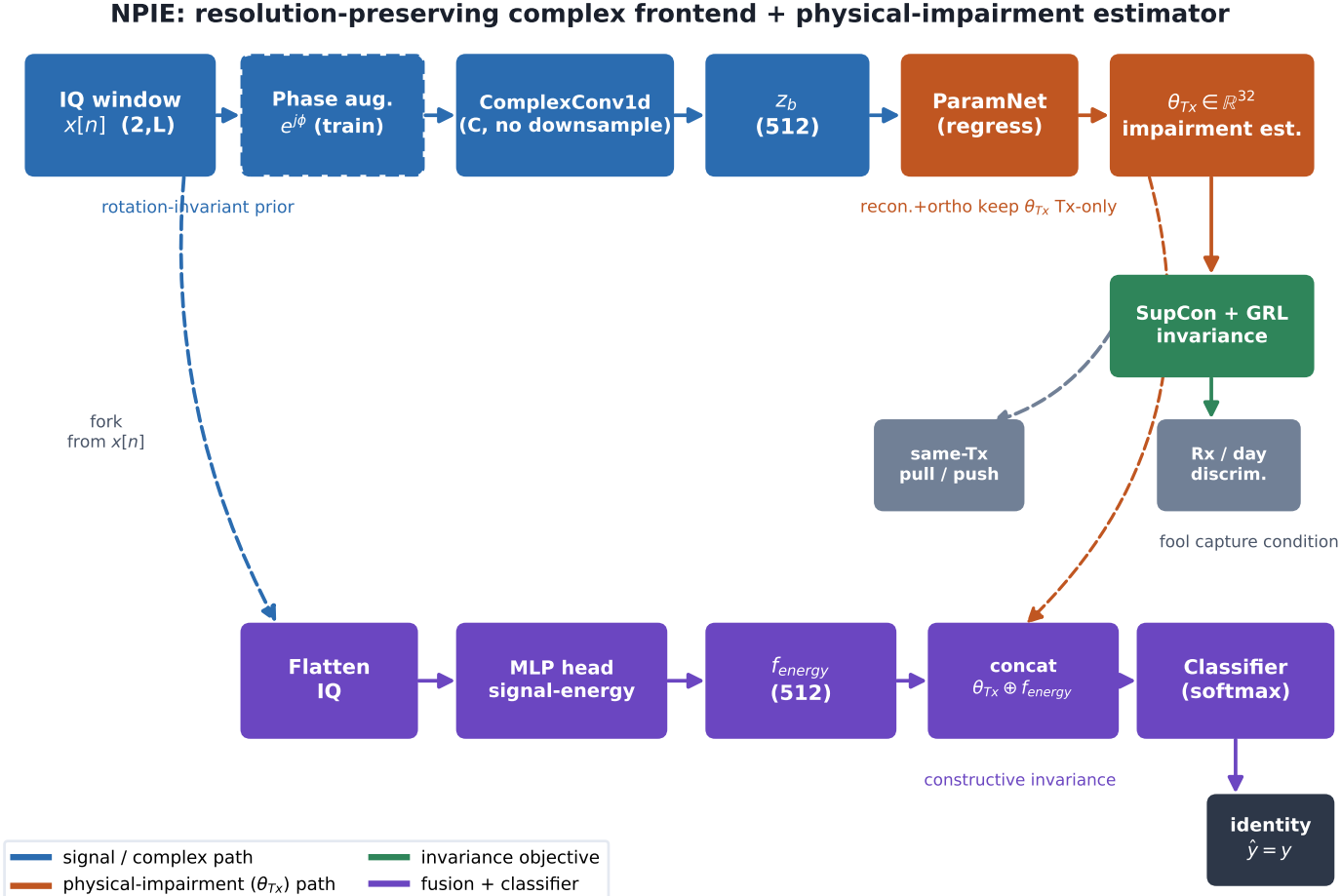
NPIE: resolution-preserving complex frontend + physical-impairment estimator
IQ window x[n] (2,L)
Phase aug. e^{jφ} (train)
ComplexConv1d (C, no downsample)
z_b (512)
ParamNet (regress)
θ_Tx ∈ ℝ^32 impairment est.
rotation-invariant prior
recon.+ortho keep θ_Tx Tx-only
SupCon + GRL invariance
same-Tx pull / push
Rx / day discrim.
fool capture condition
fork from x[n]
Flatten IQ
MLP head signal-energy
f_energy (512)
concat θ_Tx ⊕ f_energy
Classifier (softmax)
constructive invariance
identity ŷ = y
signal / complex path
physical-impairment (θ_Tx) path
invariance objective
fusion + classifier